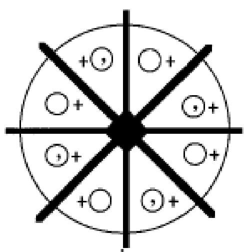
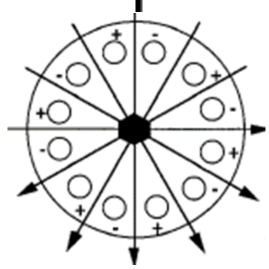
Q1) 3D Point Groups: (20 pts)

a) Using a projection sketch, determine which point group results from adding a center of symmetry to 222

b) Use the following projections to identify the point group and crystal system, describe the symmetry elements (type and orientation) and order of the group.





Point group	Crystal system	Symmetries	Group order

Q2) Point and space groups: (30 pts)

a) Using sketches, determine which point groups result from adding a center of symmetry to the following point groups: 1, m, $mm2$, $4mm$, 6, $\bar{6}$, $6mm$.

b) Which of the point groups in a) are centrosymmetric?

c) Use a diagram to determine what point group results from combination of:

   a. Two mirror planes oriented at 90° with respect to one another and a diad parallel to both planes.

   b. A tetrad axis with a perpendicular diad.

   c. A 4-bar axis with a mirror oriented with its normal perpendicular to the roto-inversion axis

   d. A triad axis with a perpendicular diad

   e. A hexad axis with a perpendicular mirror

d) Does the point group $\bar{3}m$ contain symmetry elements not present in the name?

e) Specify the point group and Bravais lattices of the crystals that have the following space groups: $P\bar{4}2_1c$, $Ccc2$, $P6_3mc$
f) Which of the space groups in e) lack centrosymmetry?

Q3) General coordinates of point groups and space groups (20 pts)

a) Using the matrix approach, determine the general coordinates of the point group 4 and the order of the point group.
b) Determine the general coordinates of the point group 4mm based on your answer in a) and the order of the point group.
c) What are the general coordinates of I4mm? What is the order of the space group?

Q4) Bravais lattices and space groups: (30 pts)

One common structure for compounds with the ABO_4 stoichiometry is Scheelite, typified by $CaWO_4$.

Table 3B.3. *The structure of scheelite, $CaWO_4$.*

Formula unit:			$CaWO_4$
Space group:			$I4_1/a$ (no. 88)
Cell dimensions:			a = 5.24 Å, c = 11.38 Å
Cell contents:			4 formula units
Atomic positions:			
	Ca	(4b)	(0,0,1/2); (1/2,0,1/4); +I
	W	(4a)	(0,0,0); (0,1/2,1/4); +I
	O	(16f)	(x, y, z); $(\bar{x}, \bar{y}, z)$; $(x, y+1/2, 1/4-z)$; $(\bar{x}, 1/2-y, 1/4-z)$; $(\bar{y}, x, \bar{z})$; $(y, \bar{x}, \bar{z})$; $(\bar{y}, x+1/2, z+1/4)$; $(y, 1/2-x, z+1/4)$; +I
			$x=0.25$, $y=0.15$, $z=0.075$

a) What is the Bravais lattice of this crystal?
b) Is scheelite centrosymmetric or non-centrosymmetric?
c) Sketch a projection down [100]
d) Describe the coordination of the W atoms and calculate the bond distances
e) Are the general positions of this group occupied?
f) What is the point symmetry of the site occupied by W?
g) With respect to the atoms in the unit cell, where is the glide plane?